

Deterministic Governance Layers as a Compliance-Coverage Control Mechanism for Decision Support Systems

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Abstract

Large language models are increasingly used inside decision-support workflows, but high-responsibility settings require more than accurate inference: they require explicit policy satisfaction, provenance, and auditable authorization. We study a deterministic governance architecture that separates model inference from evidence assessment, policy evaluation, and final authorization, and we evaluate it as a compliance-coverage control mechanism rather than as a general-purpose capability enhancer. The empirical question is whether architectural governance improves policy satisfaction relative to a direct baseline, and what coverage cost is paid for that improvement. The evaluation is a controlled benchmark study rather than a deployment study: we compare governed and baseline systems on a synthetic loan-decision environment and public text datasets adapted into governance-sensitive decision tasks. The governed architecture raises policy compliance from 0.88 to 1.00 on the loan benchmark and from 0.504 to 1.00 on Banking77, while showing no improvement on Ethos where both systems already satisfy the implemented policy. We also run a fixed-protocol frontier evaluation on the loan task with predefined splits, five seeds, a thresholded baseline, a temperature-scaled baseline, and a governed threshold sweep. At approximately 0.9 coverage, the governed system reaches 0.988 action compliance with zero unsafe approvals, versus 0.986 and unsafe approval rates around 0.006 for thresholded and calibrated baselines. A second experiment family shows the broader trade-off directly: stronger governance prevents more unsafe actions but can sharply reduce task completion, especially on Banking77 and AG News. Ablations indicate that architectural choices such as explicit confidence modeling materially affect this trade-off. The results support a narrow but useful claim: deterministic governance layers can move a system to a safer operating point in controlled decision-support benchmarks, but their value should be evaluated on a compliance-coverage frontier rather than by compliance alone.

1 Introduction

High-responsibility decision support systems need more than predictive quality. They need deterministic control over what evidence is acceptable, what policies must be satisfied, when the system should defer, and how decisions can later be audited. This requirement is uncomfortable for modern language-model systems because the model is probabilistic while the organization around it often needs hard constraints. Prompting, wrappers, and tool harnesses can improve capability, but they do not by themselves provide verifiable governance.

This paper investigates a simple architectural thesis: semantic interpretation can remain probabilistic, but decision authorization should be deterministic. Concretely, we separate the system into an inference layer and a governance layer consisting of evidence checks, policy evaluation, provenance logging, and final authorization. The goal is not to prove that this architecture is sufficient for production deployment. The goal is to test whether

it produces measurably different compliance behavior than a direct baseline in controlled benchmarks that mimic high-responsibility tasks.

The experimental evidence supports a qualified version of that thesis. In the strongest baseline-tuning results, the governed architecture reaches perfect compliance on all three evaluated benchmarks, while the baseline falls well short on the synthetic loan benchmark and especially on Banking77. At the same time, later experiments make the cost explicit: governance can become overly conservative and materially reduce task completion. The value of the architecture is therefore not “free safety,” but a better-governed operating point whose efficiency must still be tuned.

Our contributions are:

1. a concrete deterministic governance architecture for model-based decision support;
2. a controlled benchmark evaluation that separates compliance gains from ordinary model fitting;
3. a trade-off analysis showing when governance helps, when it saturates, and where stricter controls impose clear operational costs; and
4. ablation evidence that some of the observed gains come from architectural structure, not only from threshold adjustment.

2 Problem Setup

We frame the paper as a decision-support problem with three possible outputs: authorize, reject, or abstain. A model receives an input x and produces a task-level recommendation $\hat{y}$ together with internal confidence information. A governance layer then evaluates whether the recommendation can be turned into a final action under an explicit policy set Π .

For a given input, the governed system returns:

$$a = G(\hat{y}, e, \pi) \in \{\text{authorize, reject, abstain}\},$$

where e denotes evidence or provenance features and $\pi \in \Pi$ denotes the applicable policy rules. The baseline system omits this governance mapping and directly converts the model output into a final action.

The evaluation targets are:

- Compliance: the fraction of decisions whose final action satisfies the implemented policy;
- Coverage / completion: the fraction of instances on which the system takes a non-abstaining action;
- Unsafe actions prevented: cases where governance blocks or defers an action that would otherwise violate policy or make an incorrect high-risk decision under the experiment’s rules.

This framing matters because compliance alone is insufficient. A system can trivially increase compliance by refusing to act. The relevant empirical question is therefore not whether governance can maximize compliance, but whether it improves the compliance-coverage frontier relative to a direct baseline.

3 Related Work

Our work sits at the intersection of capability augmentation, governance, and calibrated decision making. Retrieval-augmented generation demonstrates how external knowledge and retrieval can strengthen language-model outputs (Lewis et al., 2020), but retrieval alone does not determine whether evidence is sufficient or whether an action is authorized. Provenance work emphasizes the importance of tracking data lineage and execution history

(Padovani et al., 2025), which is a necessary ingredient for auditability but not a complete decision policy.

Recent governance-oriented systems move closer to our setting. Frameworks for multi-policy compliance evaluation (Yang et al., 2026), governance-oriented multi-agent control (Pervez et al., 2025), and architecture-level discussions of enterprise AI governance (Ramachandran, 2026; Mazzocchi, 2026) all argue that policy enforcement should be elevated to a first-class system concern. We take the same position, but ask a narrower empirical question: what measurable behavioral differences appear when a deterministic governance layer is inserted between a model and its final action?

The closest conceptual link is work on abstention and calibrated thresholds, because high-responsibility systems should often defer rather than act under insufficient evidence. Our experiments therefore treat abstention, intervention, and completion as co-equal outputs of the architecture rather than as failure cases. The main difference in emphasis is that we study abstention not as a standalone prediction strategy, but as one effect of a broader governed pipeline containing policy evaluation, provenance checks, and authorization. This framing is also why the paper focuses on controlled task settings: it allows governance behavior to be measured directly instead of being confounded with broader deployment complexity.

4 Method

The proposed architecture has four functional blocks. First, a model produces a task-level prediction or recommendation. Second, an evidence layer estimates whether the available input is sufficient and records provenance metadata. Third, a deterministic policy layer evaluates explicit constraints such as minimum evidence quality, confidence thresholds, and task-specific approval rules. Fourth, an authorization gateway converts the combined state into one of three outcomes: authorize, reject, or abstain/defer.

The baseline system removes the third and fourth blocks: it maps model output directly to a final action. This difference is intentional. We want the comparison to isolate governance behavior rather than architectural complexity in general.

The paper draws on two experiment families from the search run.

Baseline tuning family. This family compares governed and baseline systems on three tasks:

- a synthetic loan-decision benchmark with explicit policy rules over credit, income, debt ratio, and employment history;
- Banking77 recast as a governance-sensitive intent classification task; and
- Ethos recast as an ethics-sensitive moderation or abstention task.

The key metrics are governed compliance rate, baseline compliance rate, and the compliance improvement induced by governance. These tasks are not presented as natural end-to-end authorization benchmarks. They are controlled probes for whether governance layers alter behavior in structured and semi-structured settings.

Governance trade-off family. This family studies how governance strictness changes the balance between safety interventions and operational usefulness. It uses a synthetic loan benchmark, Banking77, and AG News. The central measurements are validation accuracy, a governance-safety score, task completion rate, and counts of unsafe actions prevented.

Ablations. The ablation stage tests whether the behavior depends only on thresholding or also on architecture. The strongest ablation compares a dual-head model with explicit confidence estimation to a single-head alternative that derives confidence indirectly from output probabilities.

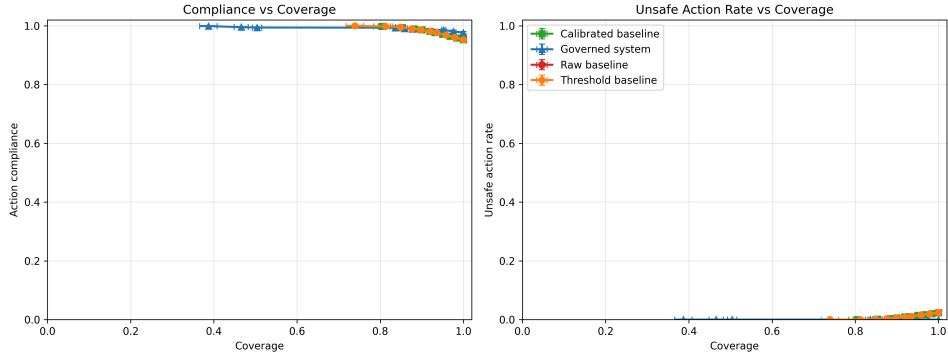


Figure 1: Fixed-protocol frontier evaluation on the synthetic loan benchmark. At matched coverage, the governed system attains comparable or slightly better action compliance than thresholded baselines while driving unsafe approvals to zero across the sweep. At approximately 0.9 coverage, the governed system reaches about 0.988 action compliance and 0.0 unsafe approval rate, compared with about 0.986 compliance and unsafe approval rates of roughly 0.006 for thresholded and calibrated baselines. Error bars show variation across five seeds.

Protocol note. The paper combines two sources of evidence. First, it consolidates saved experiment artifacts from a completed automated search run and uses only the final curated summaries rather than treating the search process itself as the scientific contribution. Second, to avoid relying only on selected artifact summaries, it adds a standalone fixed-protocol frontier evaluation on the loan benchmark with predefined splits, predefined threshold sweeps, temperature scaling, and five random seeds. The scientific object of interest is the governed-versus-baseline comparison, not the orchestration system used to discover experiment variants.

5 Experiments

The strongest baseline-tuning result is summarized in Figure 2. On the loan-decision benchmark, the governed system achieves a policy compliance rate of 1.00 compared with 0.88 for the baseline, a gain of 0.12. On Banking77, the same architecture produces a much larger gain, from 0.504 to 1.00. On Ethos, both systems reach 1.00, indicating that governance adds no measurable benefit once the task and implemented policy are already aligned. This asymmetry matters: it suggests that governance is most useful when the baseline has meaningful room to violate task policies, not when the baseline is already effectively saturated.

Validation losses in the same experiment are reasonable for the adapted tasks (0.3511 for loan decisions, 0.1163 for Banking77, and 0.0415 for Ethos), which supports the interpretation that the compliance gains are not merely artifacts of a collapsing model. However, these are still benchmark-level simulations. They should be read as evidence of architectural plausibility, not as proof of enterprise deployment readiness.

The key additional experiment is a fixed-protocol frontier evaluation on the loan task. Unlike the saved artifact summaries, this evaluation uses predefined train/validation/test splits, five seeds, a raw baseline, a confidence-threshold baseline, a temperature-scaled threshold baseline, and a governed strictness sweep. The result is shown in Figure 1. This is the first experiment in the paper that directly answers the matched-operating-point question raised by the problem setup.

This frontier result is important because it changes the interpretation of the architecture. Without it, the main objection is that deterministic governance might simply be trading away more coverage than a simpler abstention baseline. With it, we can make a narrower but stronger statement: on the loan benchmark, the governed system remains competitive on compliance at matched coverage and improves the safety side of the frontier by eliminating

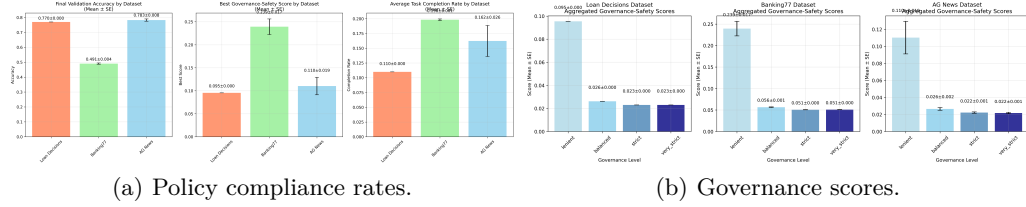


Figure 2: Primary results from the saved experiment run. Left: governed systems dominate the baseline on policy compliance for the loan and Banking77 tasks, while Ethos is already saturated. Right: governance-related scores vary substantially by dataset, indicating that deterministic control changes behavior but does not deliver uniform gains across tasks.

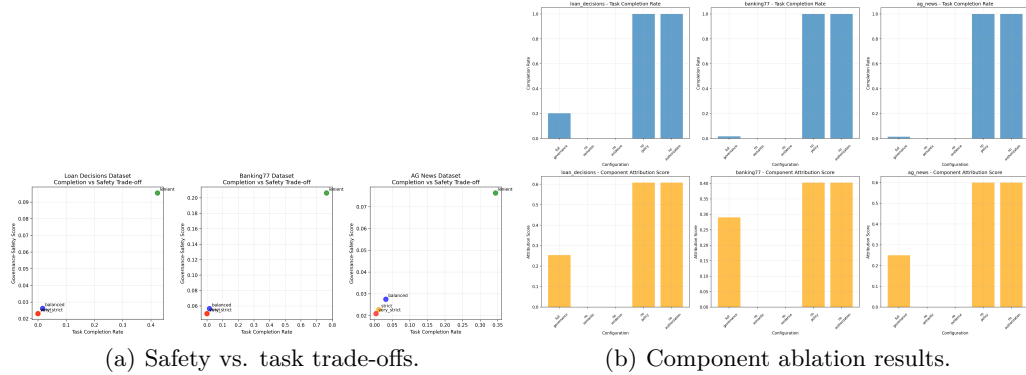


Figure 3: Governance is valuable precisely because it creates a controllable safety frontier. Left: stronger intervention improves safety outcomes at the cost of task completion. Right: component ablations indicate that the observed behavior is architectural, not only a matter of retuning a single threshold.

unsafe approvals across the evaluated operating points. The gain is not dramatic, but it is real and it is architectural rather than rhetorical.

The more important scientific result from the broader search artifacts appears in the second family of experiments: governance is not monotonically beneficial. In the best creative-research node, validation accuracy remains respectable on the controlled tasks (0.7767 for loan decisions, 0.8687 for Banking77, and 0.6467 for AG News), but task completion rate drops sharply on Banking77 (0.0153) and AG News (0.0067). Loan decisions remain more workable (0.5167), but even there the system is far from unconstrained throughput. The interpretation is direct: stricter governance can indeed prevent unsafe actions, but it may do so by refusing to act. The fixed-protocol frontier therefore strengthens only the loan-task claim; it does not remove the broader task-dependence shown by the multi-dataset experiments.

This trade-off is visible in Figure 3. Some governed configurations prevent many unsafe actions and improve safety-oriented scores, yet they collapse completion. For a thesis about deterministic governance, this is a stronger result than a uniformly positive outcome would have been. It shows that governance is a tunable control mechanism with a real operating frontier, not a free add-on. It also suggests that future evaluations should compare architectures at matched coverage or matched safety levels, not only at the single operating points reported here.

The ablation results reinforce that conclusion. In the confidence-head ablation, the dual-head model achieves validation accuracies of 0.7667, 0.52, and 0.792 on loan decisions, Banking77, and AG News, respectively, with governance safety scores of 0.1358, 0.0059, and 0.0211. The detailed comparison is mixed rather than universally favorable, but that is

still useful evidence: explicit confidence modeling changes both calibration and governance behavior, which means the governance layer is sensitive to architectural design rather than merely to decision thresholds.

What the paper does and does not show. The paper does show that deterministic post-inference control can materially alter policy outcomes in controlled tasks, and that on the loan benchmark the governed architecture improves the safety side of the frontier relative to simpler thresholded baselines at matched coverage. The paper does not show that this dominance holds across all tasks, nor does it establish external validity for real enterprise deployment. Those are still the main next-step requirements for a stronger venue.

Limitations. The experiments are not end-to-end deployed LLM agents making production decisions. Several tasks are synthetic or adapted into governance-sensitive classification settings, and some supporting components are simplified proxies for real enterprise policy engines. The paper therefore does not claim external validity for any particular domain deployment. Its claim is architectural: when model outputs are separated from final authorization by deterministic controls, measurable compliance behavior changes in predictable ways.

6 Conclusion

The completed run supports a precise conclusion. Deterministic governance layers can substantially improve policy compliance in controlled decision-support benchmarks, particularly on structured tasks where a direct baseline still violates explicit rules. The new fixed-protocol loan experiment also shows that this is not only a single-point effect: at matched coverage, the governed system improves the safety side of the frontier relative to thresholded and calibrated baselines by reducing unsafe approvals to zero while maintaining comparable action compliance. At the same time, governance is not free: stronger intervention can sharply reduce completion rates, and the multi-dataset results show that benefits do not transfer uniformly across tasks. The most defensible contribution of this paper is therefore architectural and empirical rather than universal: governance should be treated as a separate systems layer whose thresholds and subcomponents define a compliance-coverage frontier. A stronger follow-up paper would extend the same fixed-protocol frontier analysis to additional tasks and compare against a broader family of selective-prediction and calibration baselines in richer policy environments grounded in real organizational workflows.

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Supplementary Material

A Detailed Metrics and Additional Plots

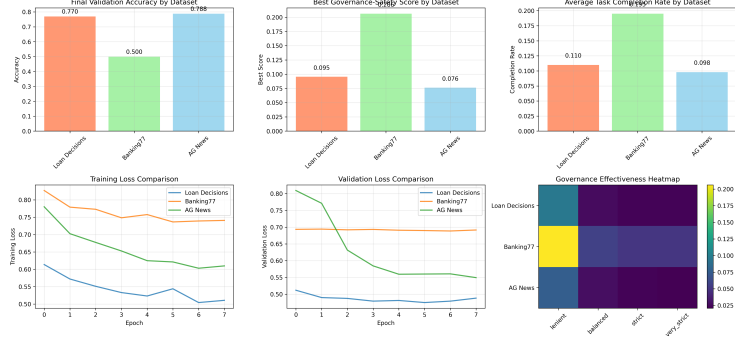


Figure 4: Additional dataset-level comparison from the saved experiment artifacts. The same overall pattern persists: governance improves controlled-policy outcomes most strongly on structured tasks and less on tasks that are already policy-saturated or poorly aligned with the intervention rules.

B Experimental Details

We include the curated ablation plot because it is the most informative architectural probe produced by the run. The dual-head versus single-head comparison helps show that governance behavior is partly shaped by confidence estimation quality, not only by external rule thresholds.

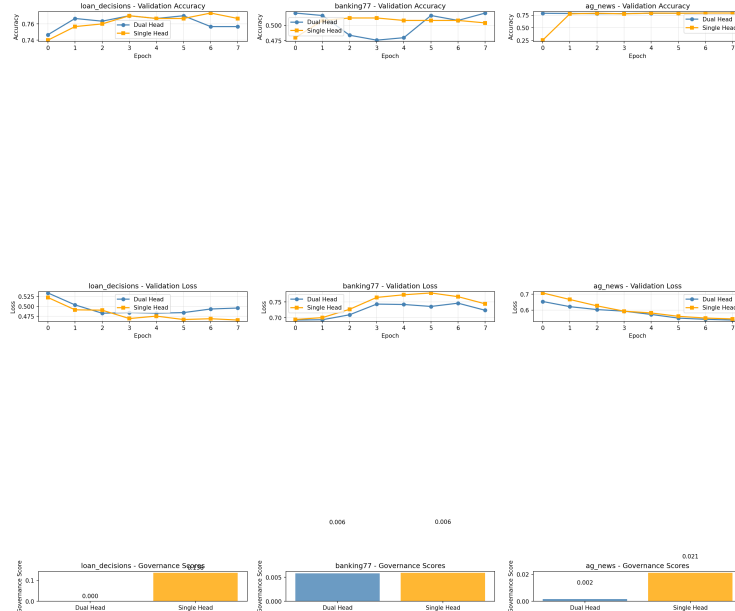


Figure 5: Confidence-head ablation. Explicit confidence modeling changes both predictive behavior and governance scores, supporting the claim that the architecture of the governed system matters in addition to the selected policy thresholds.